Md Sadat Tanzim

Mechanical Engineering · Robotics and Autonomous Systems

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CONTACT

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TECHNICAL SKILLS

Atomistic Simulation
LAMMPS (MD, hybrid MD/MC), OVITO

CFD and FEA
ANSYS Fluent, ANSYS Static Structural, SolidWorks Simulation

Design
SolidWorks, Altium, PCB design and fabrication

Programming
Python, C/C++ (Arduino), LaTeX

Robotics and Embedded
OpenCV, MAVSDK, Raspberry Pi, Arduino, ESP32, sensor integration

AWARDS AND HONOURS

1st Runner-Up
Project Showcasing (DroverNet), EEE Day 2026, Department of EEE, BUET — approx. 30 competing teams

President's Scout Award
2017. Highest scouting award in Bangladesh, conferred by the President of the People's Republic of Bangladesh

LEADERSHIP

Vice President
BUET Robotics Society
2026 – Present

Project Manager
Team NUVOLA — competitive robotics team under BUET Robotics Society
2026 – Present

President
BADHON, Kazi Nazrul Islam Hall Unit — voluntary blood donation organisation across public universities in Bangladesh
2026 – Present

TEACHING

Instructor, Basic Robotics Workshop
BUET Robotics Society, 2026
Introductory robotics for junior undergraduates: microcontroller programming, sensor interfacing, actuator control.

RESEARCH INTERESTS

Robotics and autonomous systems, with particular interest in perception-driven navigation, multi-agent aerial–ground coordination, and low-cost autonomy for unstructured environments. Secondary interest in atomistic simulation of structural materials, specifically the interaction between chemical short-range order and deformation mechanisms in nanostructured alloys.

EDUCATION

B.Sc. in Mechanical Engineering *November 2022 – July 2027 (expected)*
Bangladesh University of Engineering and Technology (BUET), Dhaka, Bangladesh
Currently Level 4, Term 1 · CGPA: 3.25 / 4.00
Thesis: *Chemical Short-Range Order in Nanotwinned CoCrNi Nanowires: Distribution Across Surfaces and Twin Boundaries and Its Effect on Yield Strength*
Supervisor: Professor Dr. Md. Afsar Ali, Department of Mechanical Engineering, BUET

Higher Secondary Certificate (HSC) *2021*
Satkhira Government College · GPA: 5.00 / 5.00

Secondary School Certificate (SSC) *2019*
Satkhira Government High School · GPA: 5.00 / 5.00

PUBLICATIONS

A. S. Tahmid, **M. S. Tanzim**, A. R. Junaid, and T. A. Tonoy, "Aerial Perception-Driven Navigation for Low-Cost Ground Vehicles in GPS-Denied Scenario," *IEEE MARSCON 2026*, Ahsanullah University of Science and Technology, Dhaka, Bangladesh, 17 August 2026.

All authors contributed equally; presented by A. S. Tahmid.

RESEARCH EXPERIENCE

Undergraduate Thesis Researcher *January 2026 – Present*
Department of Mechanical Engineering, BUET · Supervisor: Dr. Md. Afsar Ali

- Simulating free-standing nanotwinned CoCrNi nanowires in LAMMPS across four diameters (5–16 nm) and four twin conditions; ordering generated by hybrid MD/MC annealing with swap attempts ocalized per atom and quantified by Warren-Cowley parameters referenced against composition-preserving null models.
- Resolved the ordering budget as approximately 91% core, 9% twin boundaries, and none at the free surface, and located 82–93% of first plastic events at the surface — evidence that CSRO strengthens by raising surface nucleation barriers rather than by bulk glide resistance.
- Measured twin-boundary strengthening at ≈1.07 Gpa in the random state against ≈0.48 Gpa in the ordered state, establishing the two mechanisms as non-additive, alongside an inverted size effect in which larger diameters prove stronger.

Hardware Lead *October 2025 – August 2026*
Collaborative Aerial–Ground Autonomous Navigation System, BUET

- Built the full hardware stack for a ocalizedzed UAV–UGV system operating without GPS or external computation: Raspberry Pi 5 aerial unit, Arduino Mega 2560 skid-drive ground unit, and a WebSocket-to-ESP32-to-UART command chain that held data integrity across every field trial.
- Calibrated three AS5600 12-bit magnetic encoders through a TCA9548A multiplexer, deriving a linear correction model from systematic rotation trials that eliminated consistent motor undershoot ($R^2 \geq 0.96$ across four motors).
- Field trials over a 9 m² arena at 10 m hover altitude: 10–20 cm positional accuracy on paths under 3 m, 1–2 s from image capture to planned path, and 2–3 minute end-to-end missions.

Electrical Subsystem Lead *Proposal under review*
Lower-Tropospheric $PM_{2.5}$ and CO_2 Profiling CanSat · Submitted to Bangladesh Space Research and Remote Sensing Organization (SPARRSO)

- Leading the electrical subsystem: PCB design and fabrication, LiPo main pack with backup-cell power architecture, thermostat-controlled heater circuitry for −50 to −60 °C stratospheric operation, and sensor, motor, and ESC integration.
- Payload carries PMS5003 particulate, MH-Z19B NDIR CO_2, BME280, and NEO-6M GPS sensing

INDUSTRIAL EXPERIENCE

Industrial Attachment Intern
MGI Sugar Mill, Bangladesh
July – August 2026
Mandatory curriculum placement. Followed the full production line from raw cane intake through refining to packaging, with emphasis on in-process quality control.

to approximately 30 km on a latex balloon, with GPS/barometer-triggered active descent below 3000 m over a mandatory parachute fallback.

- Co-authored a 10-month work plan and BDT 192,940 budget targeting a pilot 0–3 km $PM_{2.5}$ and CO_2 profile over Bangladesh, addressing a gap in publicly available vertical air-quality data.

ACADEMIC PROJECTS

Swirl-Induced Turbulence in a Double Pipe Heat Exchanger *2025*
ME 310: Heat Transfer Equipment Design

- Designed, fabricated, and analysed a 1.016 m AISI 304 stainless steel double pipe exchanger with a full-length twisted-tape swirl insert, operating at Re 9332 in the inner tube and Re 5894 in the annulus.
- ANSYS Fluent analysis showed the swirl configuration raising heat transfer from approximately 800 W to 1350 W (+68.7%) and dropping hot-stream outlet temperature from 66.9 °C to 52.1 °C, at the cost of elevated pressure drop; energy imbalance held under 2% in both cases.
- Prototype sustained hydraulic testing at 150 psi inner tube and 100 psi annulus with zero leakage across all joints and welds.

Design and Analysis of an Ackermann Steering Mechanism *2025*
ME 352: Machine Design Sessional

- Verified the Ackermann kinematic condition analytically for a 1500 mm track and 2500 mm wheelbase (23.2° outer against 30° inner) and computed 27.30 Mpa Lewis bending stress on a 21-tooth pinion under 15 N·m, a factor of safety of roughly 8 against 220.6 Mpa yield.
- Diagnosed the divergence between theory and FEA (9,704 Mpa on the gear, 15,328 Mpa on the U-joint in SolidWorks and ANSYS) as contact-patch stress singularity and unconstrained rigid-body motion rather than physical failure, and specified ocalized mesh refinement with non-linear contact formulation as the fix.